Lab 4 plus

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

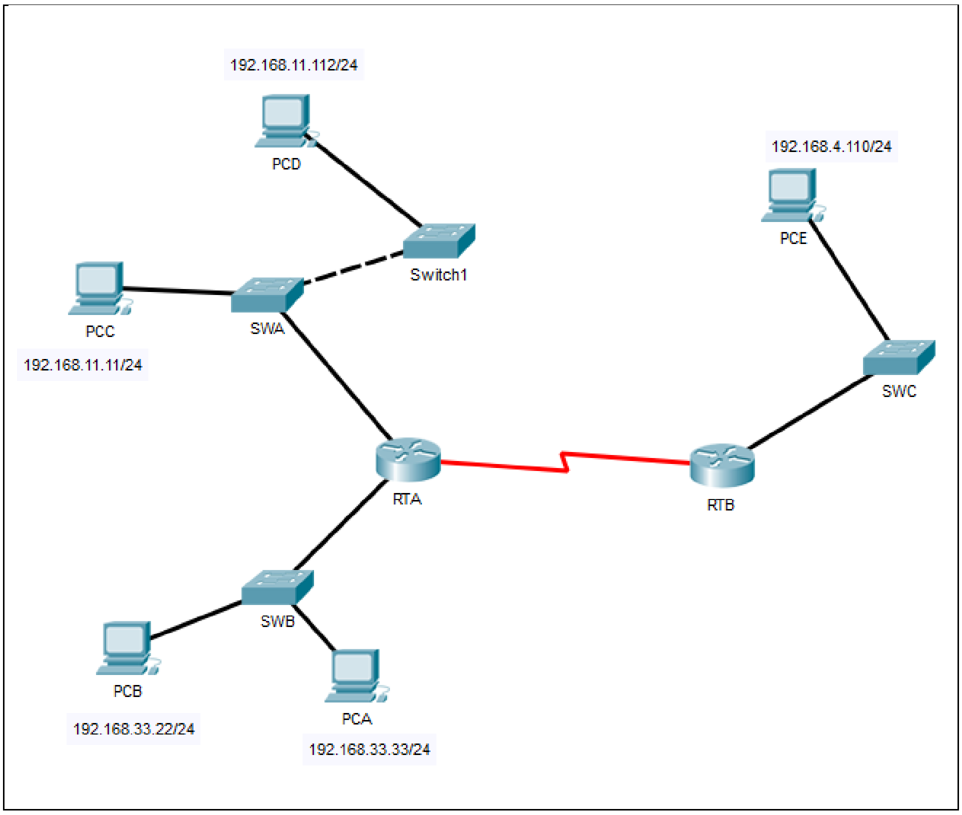


Figure 1

Part 1: Review the topology

Step 1: Open the ***Lab_4_ARP.pkt***, and Perform the following tasks.

a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1            -   0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1            -   000C.CF0C.593A  ARPA   FastEthernet0/0
```

b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1             -   0001.977A.B614  ARPA   FastEthernet0/0
```

c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

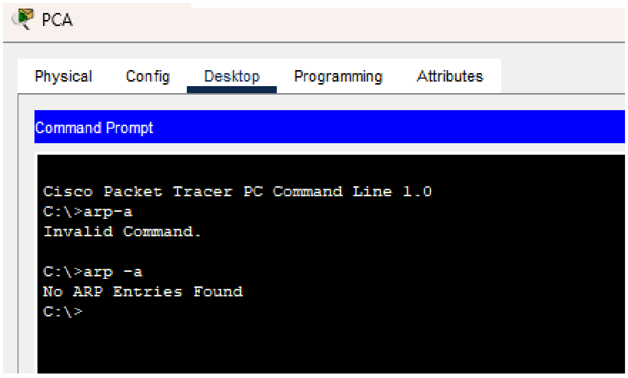
```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

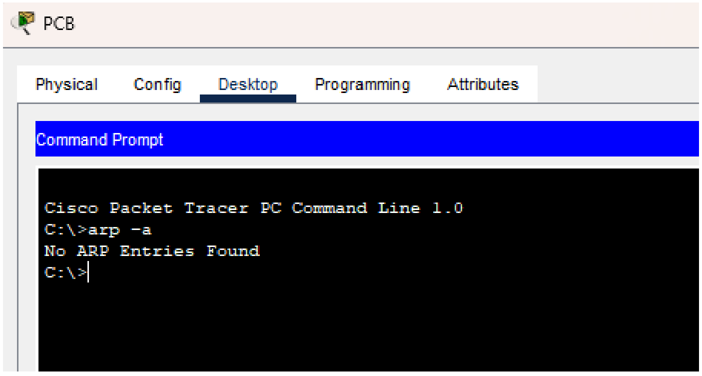
```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
```

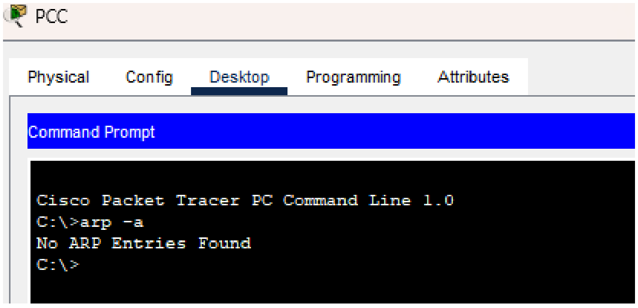
```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
```

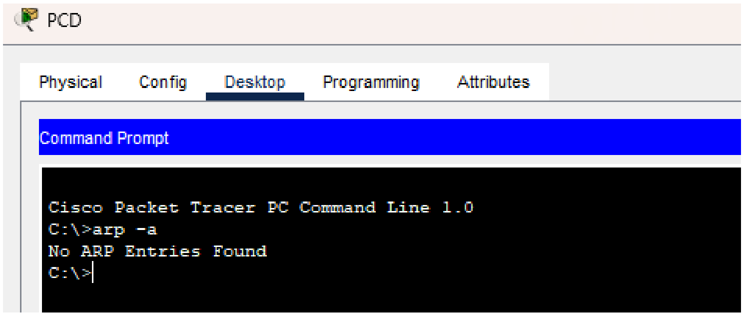
```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
```

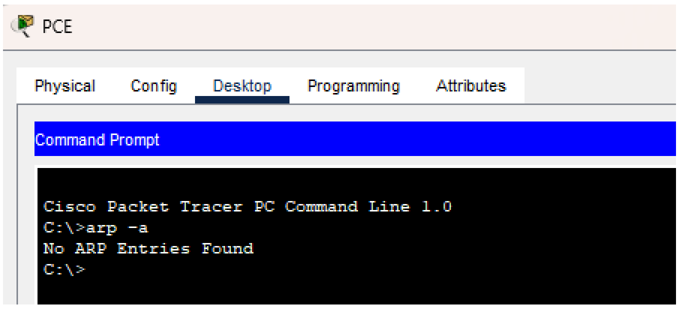
d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.











e. What are your thoughts on the results?

I think when we type arp -a and it shows "No ARP entries found," it means the ARP table is empty. This happens because the PC hasn't communicated with any devices recently, no ARP requests have been made, or the ARP entries have expired.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

a. Click **PCA** and click the Desktop tab > Command Prompt.
b. Enter the **ping 192.168.33.22** command. This may take a few seconds.
c. In the Command prompt of PCA, type **arp –a**. Paste the result of this command here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time=1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.22         0060.47ea.a746        dynamic
```

d. In the Command prompt of PCB, type **arp –a**. Paste the result of this command here

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.33.33         0002.1755.9a06        dynamic
```

e. In the Command prompt of PCC, PCD and PCE, type **arp –a**. Paste the result of this command here.

PCC:

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCD:

```
C:\>arp -a
No ARP Entries Found
C:\>
```

PCE:

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Step 2: Generate traffic between PCC to all other PC except PCA.

a. Click **PCC** and click the Desktop tab > Command Prompt.

b. Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Request timed out.
Reply from 192.168.33.22: bytes=32 time=10ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127
Reply from 192.168.33.22: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
```

c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.11.112

Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time=33ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 33ms, Average = 8ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic
```

d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp –a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Request timed out.
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=4ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.11.1          0002.4a00.0e91        dynamic
  192.168.11.112        0001.6462.0278        dynamic
```

e. Discuss the results you got from all the commands on PCC.

- **Packet Loss**:
  - The packet loss for 192.168.33.22 and 192.168.4.110 shows there might be some issues with the connection, like the network being busy, the devices not responding properly, or a routing problem.
- **Different Subnets**:
  - The ARP table doesn’t show 192.168.33.22 or 192.168.4.110, which might mean these IPs are in another subnet. That could explain why they sometimes respond to pings but don’t show up in the ARP list.
- **Good Connection**:
  - The IP 192.168.11.112 works well with no packet loss, which means it is set up correctly and connected properly.

f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.11.1            -   0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.11.11          10   00D0.D39A.C0D9  ARPA   FastEthernet1/0
Internet  192.168.33.1            -   000C.CF0C.593A  ARPA   FastEthernet0/0
Internet  192.168.33.22          10   0060.47EA.A746  ARPA   FastEthernet0/0
```

g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.4.1             -   0001.977A.B614  ARPA   FastEthernet0/0
Internet  192.168.4.110           9   0060.702D.7C08  ARPA   FastEthernet0/0
```

Step 3: Switch MAC address table.

a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0002.4a00.0e91    DYNAMIC     Fa0/1
   1    000c.8546.7d85    DYNAMIC     Fa1/1
```

b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    000c.cf0c.593a    DYNAMIC     Fa0/1
```

c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.977a.b614    DYNAMIC     Fa0/1
```

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0005.5e95.187b    DYNAMIC     Fa0/1
```

d. Do switches use arp table? (Y/N)

Yes

e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches don’t usually use ARP tables like routers do because they work differently. Regular switches, called Layer 2 switches, don’t deal with IP addresses. Instead, they use something called a MAC address table to figure out which device is connected to which port. This helps them send data to the right place within the same network.

However, if a switch has extra features like a management interface (where you can log in using an IP address to control it), it might have an ARP table, but only for its own use. This ARP table would just help the switch communicate for management purposes and wouldn’t affect how it forwards data.

So, in short, switches don’t usually need ARP tables unless they’re managing themselves using IP addresses.

f. What information does the command ***`show mac-address-table`*** gives?

Vlan, Mac Address, Type, Ports

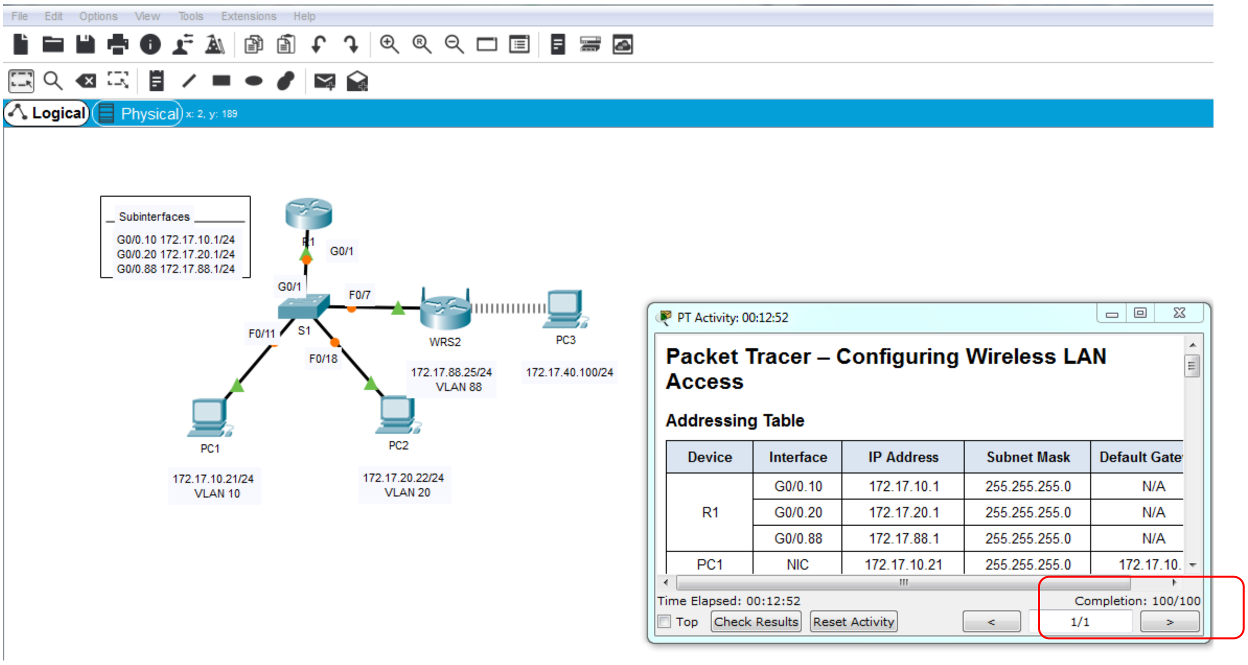
Part 3: Attach wireless lab results.

In this part, you will use ***Lab_4_Wireless.pka*** file.

Step1: Change the filename of the pka file.

a. Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*

b. Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.



c. Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.